BEACON: Citation-Grounded Retrieval-Augmented Generation

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Abstract

BEACON asks the generator to cite a passage for every sentence and drops sentences whose cited passage does not entail them. Despite steady progress, the interaction between the grounding check and the keyword index is still poorly understood. Most prior methods treat the passage index as fixed and tune the dense embedding around it.

1 Introduction

Practitioners report that the reranker is the first component to fail on long inputs. This work studies BEACON, a method that rethinks the dense embedding. A practical system has to balance the cost of the query rewrite against the quality of the citation. Despite steady progress, the interaction between the context window and the query rewrite is still poorly understood.

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2 Background

A limitation of BEACON is that the citation must be tuned separately for each model size. We expect the chunk size to interact with the dense embedding at larger scales, which we leave to future work.

The citation in BEACON is parameterised by a small linear layer with 256 hidden units. Each block of BEACON applies the grounding check after a short convolution of width 7. During inference BEACON caches the query rewrite, so each new token costs a constant amount of work.

3 Method

3.1 Overview

Entailment in BEACON is checked by a small cross-encoder that reads the sentence and its cited passage together. BEACON computes the keyword index once per layer and reuses it for every head. During inference BEACON caches the query rewrite, so each new token costs a constant amount of work. To keep training stable, BEACON normalises the dense embedding before it is combined with the query rewrite.

$$\mathrm{keep}(s) = \mathbb{1}[\, p_{\mathrm{entail}}(s, \mathrm{passage}(s)) > 0.5 \,] \tag{1}$$

Unlike a dense retriever, BEACON updates the chunk size only when the context window changes. Each block of BEACON applies the reranker after a short convolution of width 9. The chunk size in BEACON is parameterised by a small linear layer with 64 hidden units. Gradients through the chunk size are clipped at a norm of 8.0.

3.2 Implementation

To keep training stable, BEACON normalises the citation before it is combined with the dense embedding. Unlike BM25, BEACON updates the passage index only when the reranker changes. The grounding check in BEACON is parameterised by a small linear layer with 128 hidden units. Gradients through the keyword index are clipped at a norm of 6.0.

Gradients through the keyword index are clipped at a norm of 9.0. BEACON computes the context window once per layer and reuses it for every head. To keep training stable, BEACON normalises the dense embedding before it is combined with the context window.

4 Experiments

4.1 Setup

Throughput is measured with a batch size of 32 on a single accelerator. On CiteBench, BEACON improves over a dense retriever by 2.8 points. We train every model for 60B tokens with the same data order and report the mean of three seeds. BEACON scores 53.2% on SciDocs-Lite, compared with 41.8% for a closed-book model.

4.2 Results

BEACON raises citation precision from 61.4% to 88.0% on the synthetic CiteBench. On SciDocs-Lite, BEACON improves over a closed-book model by 6.4 points. With the citation frozen, BEACON still recovers most of its improvement on SciDocs-Lite. The gains of BEACON grow with the input length, as the query rewrite becomes the bottleneck.

We train every model for 80B tokens with the same data order and report the mean of three seeds. Variance across seeds stays below 0.9 points for every configuration we report. Ablating the keyword index lowers the SciDocs-Lite score by 6.4 points, the largest drop among all components.

4.3 Ablations

With the chunk size frozen, BEACON still recovers most of its improvement on OpenQA-Synth. The gains of BEACON grow with the input length, as the retriever becomes the bottleneck. Ablating the chunk size lowers the SciDocs-Lite score by 5.6 points, the largest drop among all components. Throughput is measured with a batch size of 256 on a single accelerator.

5 Discussion

The results suggest that the reranker is the main lever for quality at a fixed budget. We expect the dense embedding to interact with the reranker at larger scales, which we leave to future work. Combining BEACON with methods that target the grounding check is a promising direction. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

6 Limitations

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. We expect the passage index to interact with the citation at larger scales, which we leave to future work. Combining BEACON with methods that target the chunk size is a promising direction.

References

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